Stage-2 Report — Prognostic Analysis, FTC Sensitivity and A4 (Analysis Plan v1.1)

Date: 24 Oct 2026 **Status:** analyses complete; numbers frozen 24 Oct 2026 (special-code counts reconciled).
Runs: all on the author's machine (pinned `requirements_v2.txt`); main bootstrap B = 500 (seed 20261024); sensitivity bootstraps B = 200.
Cohort: SEER DTC, quantitative nodal data, known cause of death, complete case: n = 30,649; 243 CSS events; EPP 24.3 → unpenalised Cox primary.

1. Apparent vs optimism-corrected performance (primary, unstratified)

Model	Uno's C τ=24	Harrell's C	tdAUC 12 m	tdAUC 24 m	Cal. slope 24 m	O:E 24 m
M0 core	0.911 → **0.909**	0.917 → 0.915	0.917 → 0.916	0.914 → 0.912	0.971 → **0.965**	1.025 → 1.029
M1 + N	0.920 → **0.918**	0.924 → 0.922	0.923 → 0.922	0.922 → 0.920	0.970 → **0.960**	1.033 → 1.038
M2 + LNR	0.911 → **0.909**	0.917 → 0.915	0.917 → 0.916	0.914 → 0.912	0.971 → **0.964**	1.027 → 1.031
M3 + LODDS	0.909 → **0.907**	0.916 → 0.913	0.915 → 0.914	0.912 → 0.910	0.972 → **0.964**	1.025 → 1.029

Apparent → corrected. Optimism is 0.001–0.002 for discrimination and ≈0.01 for the calibration slope.

2. Paired discrimination differences (optimism-corrected Uno's C, τ = 24 m)

Comparison	Role	ΔC	95% CI	Holm p	Reading
M2 LNR − M1 N	primary	−0.009	−0.019 to −0.002	0.032	lower
M3 LODDS − M1 N	primary	−0.011	−0.020 to −0.002	0.032	lower
M1 N − M0 core	secondary	+0.009	+0.0005 to +0.019	—	higher (borderline)
M2 LNR − M0 core	secondary	−0.000	−0.003 to +0.003	—	comparable
M3 LODDS − M0 core	secondary	−0.002	−0.005 to +0.001	—	comparable

Other metrics (corrected): M1 − M0 on tdAUC 24 m +0.008 (−0.001 to +0.017) and Harrell's C +0.006 (−0.001 to +0.015), both CIs include 0. M2/M3 − M1 remain lower on both metrics.

3. Calibration and DCA eligibility

- Optimism-corrected 24-month slopes 0.960–0.965; O:E 1.03–1.04. Decile plots follow the identity line across two orders of magnitude of risk; the two lowest deciles have no 24-month events.
- **Calibration is acceptable; no recalibration** (validated slope inside 0.8–1.2).
- The severe miscalibration observed previously (slopes 4.8–6.5) was not reproduced under the corrected DTC specification and appeared related to the earlier modelling choices, including fixed penalisation (λ = 0.1 in lifelines shrank coefficients heavily, e.g. age HR 1.005 vs 1.06 unpenalised), broader histology mixing, and lack of T/M adjustment. No factorial decomposition was performed, so the contribution of each factor is not quantified.
- **DCA eligible.** 10-fold cross-validated 24-month risks, thresholds 0.25–10%: all four models have positive net benefit above treat-all and treat-none across the range; differences between models are ≤0.0004 (≈4 per 10,000 patients). Exploratory; Supplementary only; not used in the Abstract or Conclusion as evidence of clinical utility; no model preferred on DCA.

4. FTC-stratified sensitivity (post hoc)

Comparison (Uno's C, corrected)	Primary	Stratified by histology
M2 − M1	−0.009 (−0.019 to −0.002)	−0.008 (−0.019 to −0.0005)
M3 − M1	−0.011 (−0.020 to −0.002)	−0.010 (−0.020 to −0.001)
M1 − M0	+0.009 (+0.0005 to +0.019)	+0.008 (−0.001 to +0.019)
M2 − M0	−0.000 (−0.003 to +0.003)	+0.000 (−0.003 to +0.003)
M3 − M0	−0.002 (−0.005 to +0.001)	−0.002 (−0.005 to +0.001)

Corrected slopes 0.963–0.968. Conclusions on LNR/LODDS do not depend on the FTC proportional-hazards violation; the N-over-core gain loses CI exclusion of 0. Stratified models are not promoted to primary.

5. Prespecified sensitivity analyses (apparent ΔC, refit bootstrap B = 200)

Analysis	n / events	M2 − M1	M3 − M1	M1 − M0	M2 − M0	M3 − M0
Unknown COD as CSS (A)	30,672 / 266	−0.008 (−0.018, −0.001)	−0.009 (−0.021, −0.001)	+0.008 (+0.001, +0.019)	−0.001 (−0.004, +0.003)	−0.002 (−0.004, +0.002)
Unknown COD censored (B)	30,672 / 243	−0.009 (−0.022, −0.001)	−0.011 (−0.024, −0.001)	+0.009 (−0.000, +0.022)	−0.000 (−0.003, +0.004)	−0.002 (−0.005, +0.001)
Examined = 90 excluded	30,397 / 238	−0.007 (−0.015, −0.002)	−0.009 (−0.017, −0.001)	+0.007 (+0.001, +0.016)	+0.000 (−0.003, +0.004)	−0.001 (−0.004, +0.002)
+ surgery type + RAI	30,649 / 243	−0.012 (−0.021, −0.004)	−0.014 (−0.024, −0.006)	+0.012 (+0.004, +0.022)	+0.001 (−0.002, +0.004)	−0.002 (−0.005, +0.001)
Age as RCS	30,649 / 243	−0.009 (−0.019, −0.002)	−0.010 (−0.020, −0.003)	+0.008 (+0.002, +0.018)	−0.001 (−0.003, +0.003)	−0.002 (−0.005, +0.001)
Nodes examined ≥5	13,764 / 177	−0.004 (−0.012, +0.004)	−0.004 (−0.012, +0.003)	+0.003 (−0.004, +0.012)	−0.001 (−0.003, +0.002)	−0.001 (−0.004, +0.003)
All histologies (reference)	31,741 / 401	−0.005 (−0.010, −0.002)	−0.006 (−0.011, −0.002)	+0.005 (+0.001, +0.010)	−0.000 (−0.002, +0.002)	−0.001 (−0.003, +0.001)
Ridge, λ by 5-fold CV	30,649 / 243	−0.009 (−0.018, −0.002)	−0.011 (−0.020, −0.003)	+0.009 (+0.001, +0.019)	+0.000 (−0.003, +0.003)	−0.002 (−0.005, +0.002)

- Ridge: cross-validated partial likelihood selected λ = 0 (M1) or 0.01 (M0, M2, M3), i.e. essentially no penalisation needed.
- All sensitivity calibration slopes 0.96–1.00.
- Treatment model: RAI HR 0.30 (0.23–0.40), reflecting confounding by indication; not interpreted causally.

6. Histology and special-code descriptives (known cause of death, survival time available)

Group	n	CSS events	24-m KM CSS risk
PTC (all)	54,515	476	0.84%
FTC (incl. oncocytic)	4,103	83	1.77%
MTC	1,038	55	6.08%
ATC	714	491	76.99%
Mixed medullary	73	3	5.25%
Other (incl. NOS)	914	234	28.24%
Nodal cohort: PTC / FTC / MTC / ATC	30,103 / 1,362 / 749 / 214	235 / 23 / 24 / 117	0.73 / 1.27 / 3.55 / 60.98%
Special nodal codes 95–97, all	840	145	19.35%*
Special nodal codes 95–97, DTC	675	56	8.90%*

*Special-code counts are taken from the full cleaned file before survival-time and cause-of-death filtering (definition: examined 95/96/97 or positive 95/97). The 24-month KM risks exclude 3 records (1 DTC) with unknown cause of death. The audit's earlier figures (817 / 656) used a narrower rule that also required the other nodal field to be <98 and are superseded.

Special-code composition: PTC 644 (52 events), ATC 84 (62), Other 41 (19), MTC 40 (8), FTC 31 (4). Excluded because the codes denote aspiration only or an unknown number of nodes, so LNR/LODDS cannot be computed; the group is high-risk, so exclusion selects a lower-risk nodal cohort (Limitation). FTC, MTC and ATC are descriptive only (FTC 23 and MTC 24 nodal-cohort events).

7. A4 — ThyroidXL age reference year

- Source: ThyroidXL paper (MICCAI 2025) — data collected at the Vietnam National Hospital of Endocrinology over two years from February 2023; no per-patient examination year released. The previous reference year 2022 predates collection.
- 34 records have birth-year-coded age (the 5 others had implausible values and were already excluded).
- **Primary now 2024** (collection midpoint; maximum error about 1 year). Sensitivities 2023, 2025 and exclusion.
- Effect: all AUCs within 0.001 of the previous values; ΔAUC +0.023/+0.022/+0.025 (LR/RF/XGB) unchanged across scenarios; operating-point finding unchanged; bridge OR 0.766 unchanged.
- Updated primary diagnostic values (2024): TIRADS 0.9098; LR 0.9324; RF 0.9318; XGB 0.9353.

8. Robust vs materially changing results

Robust (all analyses):

- LNR and LODDS add no discrimination beyond the core model (every CI includes 0).
- 24-month calibration acceptable (slopes 0.96–1.00; O:E 1.01–1.07).
- Optimism negligible.

Robust in all but one analysis:

- LNR and LODDS discriminate less than N category: CI excludes 0 in the primary, FTC-stratified, COD A/B, examined = 90, treatment, RCS, ridge and all-histology analyses; **comparable in the node-yield ≥5 subgroup** (−0.004, CIs include 0).

Not robust:

- N category over the core model: CI excludes 0 for corrected Uno's C (lower limit +0.0005) and in COD-A, examined = 90, treatment, RCS, ridge and all-histology; includes 0 for corrected tdAUC 24 m and Harrell's C, the FTC-stratified model, COD-B and node-yield ≥5. Report as small and borderline.

Changed materially from the previous manuscript: calibration (from slopes 4.8–6.5 to ≈0.96); direction of the nodal comparison (previous LNR/LODDS-favourable narrative is not supported).

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9. Proposed final prognostic message

> After adjustment for age, sex, histology, T and M category, continuous LNR and LODDS provided no incremental short-term prognostic discrimination for cancer-specific mortality in differentiated thyroid carcinoma and discriminated less well than AJCC N category, whose own incremental gain over the core model was small (optimism-corrected ΔC 0.009) and not robust across sensitivity analyses. LNR remained associated with cancer-specific mortality (HR 1.85, 1.27–2.70) without improving discrimination. All models were well calibrated for 24-month risk after internal validation.

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10. Deviation-log entries

Section	Change	Timing
3.9	Post hoc histology-stratified Cox (FTC PH violation, $p < 0.001$ in all models); sensitivity only, not primary	Identified after stage-1 diagnostics; added before stage-2 inference
A4 / 2.1	Reference year 2022 → 2024 (paper: collection from Feb 2023 over two years); sensitivities 2023, 2025, exclusion; 34 records affected	After diagnostic results; documentation-driven; effect ≤0.001 AUC
3.6–3.7	Implementation: Efron Cox engine validated against lifelines; survival time 0 → 0.5 month; paired corrected Δ CI = bootstrap-sample Δ percentile interval shifted by mean Δ optimism; Uno IPCW from evaluation data	Before stage-2 runs
3.12	Sensitivities reported as apparent ΔC with refit percentile bootstrap (B = 200), not optimism-corrected	Before stage-2 runs
3.6	Ridge sensitivity: λ by 5-fold cross-validated partial likelihood, grid 0–100	Before stage-2 runs
3.11	DCA uses 10-fold cross-validated 24-month risks	Before stage-2 runs

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11. Main vs Supplementary

Main

- Fig 1. Flow diagrams for both cohorts (special codes and no-node groups shown).
- Table 1. Characteristics of both analytic cohorts.
- Table 2. Diagnostic performance: AUC, paired ΔAUC vs TIRADS, calibration, TIRADS ≥4 operating point.
- Fig 2. Diagnostic ROC and calibration curves.
- Table 3. Prognostic models: apparent and corrected discrimination, paired ΔC, calibration.
- Fig 3. Prognostic: forest plot of paired ΔC (primary + sensitivities) and 24-m calibration plot.

Supplementary

- Label-provenance audit tables (ThyroidXL) and SEER coding audit.
- Released-label, old-feature-set and A4 sensitivities; S-D1; Set B (incorporation bias); Bethesda III.
- HR tables for M0–M3; PH tests; FTC-stratified results.
- Full sensitivity table (Section 5); ridge CV; DCA curves (Supplementary only).
- Histology and special-code descriptives.
- Age-gradient integrative comparison (strata, slopes, age × sex).
- Analysis Plan v1.1 with deviation log; TRIPOD checklist.

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12. Verification checklist (numbers to add to `verify_outputs.py`)

Check	Expected	Tolerance
ThyroidXL Set A n / malignant	4,088 / 1,231	exact
Released-label malignant	1,229	exact

Labels changed by hierarchy	4 (256, 366, 370, 509)	exact
S-D1 n / malignant / benign	796 / 704 / 92	exact
Set B in S-D1 benign	88	exact
Birth-year-coded ages	34	exact
AUC TIRADS / LR / RF / XGB (2024)	0.9098 / 0.9324 / 0.9318 / 0.9353	0.0005
ΔAUC LR / RF / XGB	0.0226 / 0.0221 / 0.0255	0.0005
Specificity at TIRADS ≥4 sensitivity: TIRADS / LR	0.576 / 0.574	0.002
SEER DTC n / CSS	58,781 / 564	exact
Special codes all / DTC (n; CSS events)	840 / 675; 145 / 56	exact
Primary prognostic n / events	30,649 / 243	exact
EPP M1	24.3	0.1
Uno's C corrected M0 / M1 / M2 / M3	0.9094 / 0.9181 / 0.9091 / 0.9071	0.001
ΔC corrected M2−M1 / M3−M1 / M1−M0	−0.0089 / −0.0110 / +0.0087	0.001
Holm p M2−M1, M3−M1	0.032	0.01
Corrected slope M0–M3	0.965 / 0.960 / 0.964 / 0.964	0.005
LNR HR (M2)	1.85 (1.27–2.70)	0.01
Bridge OR / HR per 10 y	0.766 / 2.445	0.002
Ridge λ selected	0.01 / 0 / 0.01 / 0.01	exact